1. GENERAL DESCRIPTIONS

The W25Q64JV (64M-bit) Serial Flash memory provides a storage solution for systems with limited space, pins and power. The 25Q series offers flexibility and performance well beyond ordinary Serial Flash devices. They are ideal for code shadowing to RAM, executing code directly from Dual/Quad SPI (XIP) and storing voice, text and data. The device operates on a single 2.7V to 3.6V power supply with current consumption as low as 1µA for power-down. All devices are offered in space-saving packages.

The W25Q64JV array is organized into 32,768 programmable pages of 256-bytes each. Up to 256 bytes can be programmed at a time. Pages can be erased in groups of 16 (4KB sector erase), groups of 128 (32KB block erase), groups of 256 (64KB block erase) or the entire chip (chip erase). The W25Q64JV has 2,048 erasable sectors and 64 erasable blocks respectively. The small 4KB sectors allow for greater flexibility in applications that require data and parameter storage. (See Figure 2)

The W25Q64JV support the standard Serial Peripheral Interface (SPI), Dual/Quad I/O SPI Quad Peripheral Interface (QPI) as well as Double Transfer Rate(DTR) : Serial Clock, Chip Select, Serial Data I/O0 (DI), I/O1 (DO), I/O2 (/WP), and I/O3 (/HOLD). SPI clock frequencies of up to 133MHz are supported allowing equivalent clock rates of 266MHz (133MHz x 2) for Dual I/O and 532MHz (133MHz x 4) for Quad I/O when using the Fast Read Dual/Quad I/O and QPI instructions. These 3transfer rates can outperform standard Asynchronous 8 and 16-bit Parallel Flash memories. The Continuous Read Mode allows for efficient memory access with as few as 8-clocks of instruction-overhead to read a 24-bit address, allowing true XIP (execute in place) operation.

A Hold pin, Write Protect pin and programmable write protection, with top or bottom array control, provide further control flexibility. Additionally, the device supports JEDEC standard manufacturer and device ID and SFDP Register, a 64-bit Unique Serial Number and three 256-bytes Security Registers.

2. FEATURES

- **New Family of SpiFlash Memories**
  - 64M-bit /8M-byte
  - Standard SPI: CLK, /CS, DI, DO, /WP, /Hold
  - Dual SPI: CLK, /CS, IO_0, IO_1, /WP, /Hold
  - Quad SPI: CLK, /CS, IO_0, IO_1, IO_2, IO_3
  - SPI/QPI DTR (Double Transfer Rate) Read
  - Software & Hardware Reset(1)
- **Highest Performance Serial Flash**
  - 133MHz Single, Dual/Quad SPI clocks
  - 266/532MHz equivalent Dual/Quad SPI
  - 66MB/S continuous data transfer rate
  - Min. 100K Program-Erase cycles per sector
  - More than 20-year data retention
- **Efficient "Continuous Read"**
  - Continuous Read with 8/16/32/64-Byte Wrap
  - As few as 8 clocks to address memory
  - Quad Peripheral Interface (QPI) reduces instruction overhead
  - Allows true XIP (execute in place) operation
  - Outperforms X16 Parallel Flash
- **Low Power, Wide Temperature Range**
  - Single 2.7 to 3.6V supply
  - <1µA Power-down (typ.)
  - -40°C to +85°C operating range
  - -40°C to +105°C operating range
- **Flexible Architecture with 4KB sectors**
  - Uniform Sector/Block Erase (4K/32K/64K-Byte)
  - Program 1 to 256 byte per programmable page
  - Erase/Program Suspend & Resume
- **Advanced Security Features**
  - Software and Hardware Write-Protect
  - Power Supply Lock-Down and OTP protection
  - Top/Bottom, Complement array protection
  - Individual Block/Sector array protection
  - 64-Bit Unique ID for each device
  - Discoverable Parameters (SFDP) Register
  - 3X256-Bytes Security Registers with OTP locks
  - Volatile & Non-volatile Status Register Bits
- **Space Efficient Packaging**
  - 8-pin SOIC 208-mil
  - 8-pad WSON 6x5-mm / 8x6-mm
  - 8-pad XSON 4x4-mm
  - 16-pin SOIC 300-mil (additional /RESET pin)
  - 24-ball TFBGA 8x6-mm (5x5-1 ball array)
  - Contact Winbond for KGD and other options

Note: 1. Hardware /RESET pin is only available on TFBGA or SOIC16 packages